# Chirplet Fourier Analysis Network for Cross-Scene Classification of Multisource Remote Sensing Data

Xudong Zhao, Qi Ming, Yixiao Yang, Wenshuai Hu, Wei Li, *Senior Member, IEEE*, and Ran Tao, *Senior Member, IEEE*

***Abstract*—The joint application of multisource remote sensing (MSRS) data, such as hyperspectral images (HSIs) and light detection and ranging (LiDAR), offers significant potential for accurate land-cover classification. However, the existing applications often struggle with domain shifts across scenes caused by sensor, illumination, and phase variations. Focusing on this domain adaptation problem, a chirplet Fourier analysis network (ChirpFAN) is proposed for cross-scene classification of MSRS data in this article. First, a fractional spatial–frequency–phase (FrSFP) feature extraction module, including the fractional Fourier transform (FrFT) and a learnable phase-aware weighting block, is proposed to capture multidomain features. Second, a chirplet Swin Transformer (ChirpST) block integrates a chirplet Fourier analysis (ChirpFA) layer within a Swin Transformer (ST) is designed to analyze multiscale textural and oscillatory patterns. Finally, a modality-shared network including ChirpST blocks is designed for intermodal fusion and alignment. Extensive experiments demonstrate that the ChirpFAN framework achieves state-of-the-art (SOTA) performance with 3% average improvements on three challenging cross-scene MSRS datasets. The code is released on https://github.com/xudongzhao461/.**

***Index Terms*—Chirplet Fourier analysis (ChirpFA), cross-scene classification, fractional Fourier transform (FrFT), hyperspectral image (HSI), light detection and ranging (LiDAR).**

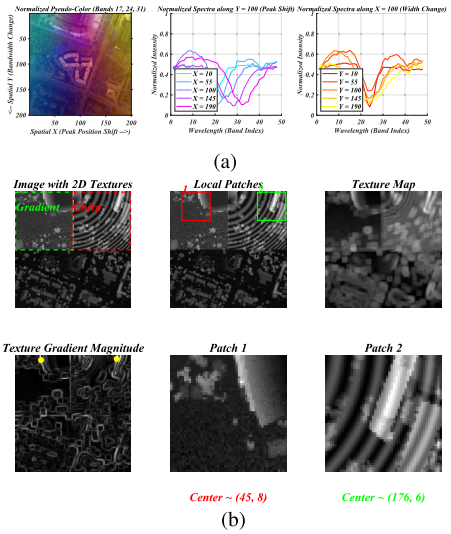



Fig. 1. HSI/LiDAR patches with nonstationary, frequency-modulated features. (a) Spectral shifts example in HSI patches. (b) Texture gradients and synthetic Chirplet dispersion in MSRS data.

## I. Introduction

RECENT progress in remote sensing technologies leads to the acquisition of large-scale and high-quality Earth observation (EO) data [1], [2]. The exponential data driven by an increasing number of satellites presents both opportunities and analytical challenges. To enhance the accuracy and robustness of land-cover classification, the focus of researchers has turned from single data sources, such as hyperspectral images (HSIs), to the integration of multisource remote sensing (MSRS) data [3], [4]. Among these sensor types, HSIs can provide detailed spectral information, while light detection and ranging (LiDAR) data offer precise 3-D structural information. The joint utilization of HSIs and LiDAR has demonstrated their complementary features and shown improvement of classification performance [5], [6].

Conventional MSRS classification methods typically operate under the assumption that training and testing samples are from the same statistical distributions and pertain to a localized scene [7], [8]. However, in practical scenarios, there are variations in sensor, light, seasonal changes, and atmospheric conditions, which lead to spectral and spatial texture discrepancies between different scenes [9], [10]. Therefore, domain shifts exist between the source data used for training and the target data where the classification models are applied [11]. Moreover, the acquisition of extensive and accurately labeled MSRS datasets for both the source and target scenes is labor-intensive and even impossible in real tasks, which degrades the generalization capability and performance.

Domain shifts in MSRS data can be seen as modulations of the standard signals due to physical processes such as atmospheric effects, illumination changes, and sensor responses. As shown in Fig. 1, spectral shifts can be seen as frequency

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modulations of spectral curves, and texture gradients include spatial modulations in textures and intensity variations. These modulations are often nonlinear and nonstationary, which vary across the images. These facts require techniques that can model such intricate and nonstationary modulations and preserve fine-grained phase information, which is a challenge for many existing domain adaptation methods [12], [13], [14].

Various domain adaptation methods are proposed focusing on domain shift [15], [16], which can be classified into three kinds; 1) statistical methods aim to minimize domain shifts by aligning statistical distributions [17], [18], [19], [20]; 2) geometric approaches include subspace alignment and optimal transport [21]; and 3) deep domain adaptation methods include adversarial training and self-supervised learning (SSL) [22], [23]. However, when applying domain adaptation methods to MSRS data, it remains challenging due to the nonlinear intermodal relations across sources [24], [25], [26], [27]. Shared-private feature alignment techniques were proposed to obtain domain-invariant features and more robust alignment [28]. However, combining sources without alignments and addressing individual shifts may increase heterogeneity, which degrades performance. Therefore, more efficient domain adaptation methods are required to effectively fuse features while addressing domain shifts between cross-scene MSRS data.

The above supervised learning methods usually depend on high-quality labeled MSRS datasets, limiting their robustness [29], [30]. In contrast, the SSL methods that use feature augmentation in transform domains have demonstrated effectiveness in utilizing the multidimensional features of labeled data while learning fine-grained features from unlabeled data [31], [32], [33]. These feature augmentation and unlabeled feature extraction methods improve the generalization by learning more intrinsic representations in the transformed domain [34]. Furthermore, recent advances in unsupervised hyperspectral classification have also shown performance in learning discriminative features from unlabeled data [35], [36], [37], [38].

The signal-level modulations of domain shift are a core physical property. While existing adaptation methods attempt to solve this problem at the statistical or geometric level, our work proposes to tackle the signal modulation directly. However, research on transform domain-based feature reconstruction and augmentation for MSRS data still confronts challenges [39], [40], [41], [42], [43]. First, conventional spatial–frequency domain analyses usually miss the phase feature, which is known to include textural and structural information. Then, it leads to the loss of details during feature reconstruction and transfer, limiting the discrimination in complex scenes. Second, the analysis of patches with small-sized local spatial–frequency windows makes it difficult to establish reliable global and local relationships for cross-scene data [44], [45]. Local windows possess a limited receptive field, which fails to incorporate contextual information to reduce feature shifts and maintain consistency. Third, it is significant to analyze the multidimensional, multimodal, and multidomain information of cross-scene MSRS data. A comprehensive understanding is essential for sufficient training networks using limited training samples.

To address these signal-level domain shifts in MSRS data, we turn to Chirplet analysis, which is designed for nonstationary signals exhibiting varying frequencies. Compared with traditional Fourier and wavelet transforms, its parametric flexibility (controlling time/frequency localization, scale, and chirp rate) allows it to create adaptive space–frequency matched filters. Domain shifts often manifest as frequency modulations in observed signals, for example, spectral features shift, and textures exhibit spatially varying frequencies due to geometric effects. Chirplets can model such linear frequency modulations, preserving phase dynamics associated with these modulations for textural and structural feature extraction. This intrinsic ability to adaptively localize and model frequency-modulated components makes Chirplets suitable to represent the complex signal distortions caused by domain shifts in MSRS data. Focusing on the above issues, a chirplet Fourier analysis network (ChirpFAN) is proposed for cross-scene classification of MSRS data. The main contributions are summarized as follows.

1) A learnable chirplet Fourier analysis (ChirpFA) method is proposed for the first time for MSRS cross-scene classification. Its properties enable an efficient and effective analysis of the nonstationary and frequency-modulated representations of MSRS textural and structural features.
2) A fractional spatial–frequency–phase (FrSFP) feature extraction module is designed to jointly analyze MSRS data. The FrSFP can extract robust and discriminative features in an optimal domain between space and frequency, while preserving phase information.
3) A chirplet Swin Transformer (ChirpST) is designed to analyze texturally rotated and frequency-modulated patterns in MSRS data. Compared with spatial–frequency analysis using fixed frequency components, it models local frequency variations and enables contexture extraction, which are robust to domain shifts.

The remainder of this article is organized as follows. Section II presents the related work. The ChirpFA theory is proposed in Section III. The proposed ChirpFAN framework is introduced in Section IV. In Section V, we present the experimental results and corresponding analysis. Finally, Section VI summarizes with some concluding remarks.

## II. Related Work

### A. Time–Frequency Analysis in Deep Learning

Deep learning has seen various integrations of time–frequency analysis, such as wavelet-based networks [46] and scattering transforms [47], which provide multiresolution analysis using fixed or learnable wavelets. However, they cannot represent signals with frequency modulations unless specialized wavelets are employed. This is a limitation when dealing with MSRS data, where domain shifts induce such complex modulations, for example, spectral warping, geometric texture distortions. Gabor filters [48] have been used in CNNs to extract textural features, but they typically rely on predefined filters or learnable filters with fixed tiling. Existing Fourier analysis networks (FANs) [49] incorporate the Fourier series to model periodicity, but focus on fixed frequencies and cannot model time-varying frequencies. The

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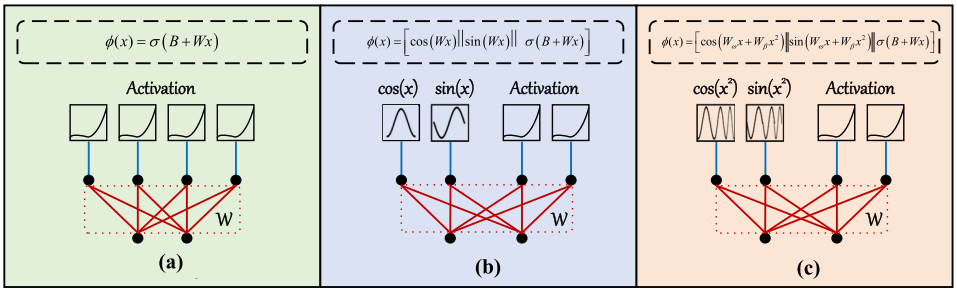



Fig. 2. Proposed ChirpFAN layer compared with MLP and FAN. (a) MLP layer. (b) FAN layer. (c) ChirpFAN layer.

proposed ChirpFA extends this by incorporating learnable chirp rates and enabling the modeling of linear frequency modulations, which are significant for representing domain shifts found in MSRS data.

### B. Fractional Fourier Analysis

The fixed-basis signal analysis approaches are unable to adapt to the diverse and localized time–frequency structures. Thus, Chirplets are designed for local adaptation by explicitly parameterizing various geometric transformations for better localized feature representation. To address the interference caused by multisensor imaging, frequency-domain methods such as the Fourier transform and the fractional transforms are utilized to capture image structures. By utilizing amplitude and phase information, the multidomain features can discriminate land covers while suppressing noise and finally provide enhanced features for the classification of MSRS data.

The eigen-decomposition definition of the fractional Fourier transform (FrFT) [50] for a signal $f(x)$ is given by the following equation:

$$\mathcal{F}_{\alpha}\left[f\right](u)=\sum_{n=0}^{\infty}c_n e^{-\mathrm{in}\alpha}\psi_n(u) \tag{1}$$

where $c_n=\int_{-\infty}^{\infty}f(x)\psi_n^*(x)dx$, $\alpha$ represents the rotation angle, and $\psi_n(x)$ are the Hermite–Gauss functions, which form an orthogonal basis.

The FrFT can also be expressed in an integral kernel form

$$\mathcal{F}_{\alpha}\left[f\right](u)=\int_{-\infty}^{\infty}K_{\alpha}(u,x)\,f(x)\,dx \tag{2}$$

where the kernel function is

$$K_{\alpha}(u,x)=\sqrt{\frac{1-i\cot\alpha}{2\pi}}\exp\left(i\pi\frac{\left(u^2+x^2\right)\cos\alpha-2ux}{\sin\alpha}\right). \tag{3}$$

For a signal $f(x)$, there exists an optimal order $\alpha_{\mathrm{opt}}$ that maximizes the energy concentration in the FrFT domain

$$\alpha_{\mathrm{opt}}=\arg\max_{\alpha}\int_{\Omega_k}\left|\mathcal{F}_{\alpha}\left[f\right](u)\right|^2du \tag{4}$$

where $\Omega_k$ represents the frequency band region containing the top $k\%$ of the energy. This theorem guarantees the convergence of learnable parameters.

### C. Fourier Analysis Networks

A multilayer perceptron (MLP) is a foundational neural network architecture [51]. As shown in Fig. 2(a), an MLP layer $\Phi(x)$ is defined as $\Phi(x)=\sigma(B_m+W_mx)$, where $\sigma$ is an activation function and $B_m$ and $W_m$ are learnable parameters (bias and weights). MLPs can approximate functions and serve as fundamental components in deep learning models. However, MLPs struggle with understanding periodicity signals.

Focusing on this problem, the FAN was proposed to efficiently model and reason about periodic signals [49]. As shown in Fig. 2(b), an FAN layer $\phi(x)$ is defined as follows:

$$\phi(x)=\left[\cos\left(W_px\right)\left\|\sin\left(W_px\right)\right\|\sigma\left(B_{\overline{p}}+W_{\overline{p}}x\right)\right] \tag{5}$$

where $W_p$, $W_{\overline{p}}$, and $B_{\overline{p}}$ are learnable parameters. The design decouples the frequency ($W_p$) and coefficient ($W_{\overline{p}}, B_{\overline{p}}$) as learning components. The FAN integrates periodicity into its structure by using the Fourier Series. It models the data periodicity rather than just memorizing patterns, which serves as a promising substitute for the MLP. Compared with the MLP, the FAN generally requires fewer parameters. However, the FAN relies on fixed frequencies $W_{\overline{p}}$, which limits it to model signals whose frequencies change over time/space, for example, domain shifts in MSRS data as shown in Fig. 1. Aiming at addressing the above problems, we propose ChirpFA in Section III by introducing learnable chirp rates.

## III. Proposed ChirpFA

The ChirpFA is proposed to extract and analyze the multidomain features. In this section, we define the ChirpFA layer, analyze its properties, and its connection to MSRS domain shifts.

### A. Definition

For a square-integrable function $f(x)\in L^2(\mathbb{R})$, it can be represented as a linear combination of basis functions. Chirplet transforms provide such a basis where functions are localized in time–frequency and can model frequency modulation [52]. For a local signal, its representation can be approximated by the following equation:

$$f(x)\approx\sum_k\left(\alpha_k\cos\left(2\pi\omega_kx+\pi\beta_kx^2\right)+\gamma_k\sin\left(2\pi\omega_kx+\pi\beta_kx^2\right)\right) \tag{6}$$

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where $\alpha_k$ and $\gamma_k$ are the linear coefficients of the $k$th Chirplet component. $\omega_k$ is the center frequency, while $\beta_k$ is the chirp rate.

Within a windowed attention framework, local image features can be interpreted as 1-D signals along a spatial dimension $x$. ChirpFAN employs chirp basis functions to model the space–frequency features of these local signals

$$\phi_{\text{chirp}}(x) = \cos\left(2\pi\omega x + \pi\beta x^2\right) \| \sin\left(2\pi\omega x + \pi\beta x^2\right) \tag{7}$$

where $\omega$ represents the initial frequency and $\beta$ represents the chirp rate. To simplify the notation for derivations, we introduce the angular frequency $\omega'_k = 2\pi\omega_k$ and the angular chirp rate $\beta'_k = \pi\beta_k$. Then, $\omega$ and $\beta$ are used to denote these simplified angular parameters.

First, we consider a network using input–output pairs $\{x_i, y_i\}$ and aim to identify a function $f(x) : \mathbb{R}^{d_x} \to \mathbb{R}^{d_y}$. A simple neural network $f_C(x)$ representing the Chirplet expansion is

$$\begin{aligned} & f_C(x) \\ & \triangleq B + \sum_{k=1}^{K} \left(\alpha_k \cos\left(\omega_k x + \beta_k x^2\right) + \gamma_k \sin\left(\omega_k x + \beta_k x^2\right)\right) \\ & = B + [\alpha_1, \ldots, \alpha_N] \cos\left([\omega_1 \| \ldots \| \omega_N]\, x + \left[\beta_1 \| \ldots \| \beta_N\right] x^2\right) \\ & + \left[\gamma_1, \ldots, \gamma_N\right] \sin\left([\omega_1 \| \ldots \| \omega_N]\, x + \left[\beta_1 \| \ldots \| \beta_N\right] x^2\right) \\ & = B + W_\alpha \cos\left(W_\omega x + W_\beta x^2\right) + W_\gamma \sin\left(W_\omega x + W_\beta x^2\right) \\ & = B + W_{\text{out}} \left[\cos\left(W_\omega x + W_\beta x^2\right) \| \sin\left(W_\omega x + W_\beta x^2\right)\right] \end{aligned} \tag{8}$$

where $B \in \mathbb{R}^{d_y}$, $W_\omega, W_\beta \in \mathbb{R}^{K \times d_x}$ represents the learnable angular frequencies and chirp rates and $W_{\text{out}} \in \mathbb{R}^{d_y \times 2K}$ are learnable parameters.

The ChirpFAN layer is designed as $\phi_{\text{chirp}}(x)$ by first decoupling $f_C(x)$ into $f_{\text{in}}(x) = [\cos(W_\omega x + W_\beta x^2) \| \sin(W_\omega x + W_\beta x^2)]$ and $f_{\text{out}}(x) = B + W_{\text{out}} x$. As shown in Fig. 2(c), the ChirpFAN layer combines this basis as follows:

$$\begin{aligned} & \phi_{\text{chirp}}(x) \\ & \triangleq \left[\cos\left(W_\omega x + W_\beta x^2\right) \| \sin\left(W_\omega x + W_\beta x^2\right) \| \sigma\left(B_{\overline{p}} + W_{\overline{p}} x\right)\right] \end{aligned} \tag{9}$$

where $W_\omega \in \mathbb{R}^{d_p \times d_x}$, $W_\beta \in \mathbb{R}^{d_p \times d_x}$, $W_{\overline{p}} \in \mathbb{R}^{d_{\overline{p}} \times d_x}$, and $B_{\overline{p}} \in \mathbb{R}^{d_{\overline{p}}}$ are learnable parameters.

### B. Properties

*1) Property:* For a 1-D signal coordinate $x$, the instantaneous frequency associated with the Chirp phase $\Phi(x) = 2\pi\omega x + \pi\beta x^2$ is

$$f_{\text{inst}}(x) = \frac{1}{2\pi} \frac{d\Phi(x)}{dx} = \omega + \beta x. \tag{10}$$

The signal energy concentrates along this linear track in the spatial–frequency plane. The Wigner–Ville distribution (WVD) [53] confirms this concentration

$$\text{WV}(x, f) = \delta(f - (\omega + \beta x)). \tag{11}$$

In contrast, FAN basis functions use fixed frequencies $\omega$, with their WVD concentrated at a single frequency line $\text{WV}(x, f) = \delta(f - \omega)$. ChirpFAN's ability to model varying frequencies provides enhanced resolution and can represent detailed textures such as edges and Chirp blurs.

*Theorem 1:* The gradient of a loss function $\mathcal{L}$ with respect to a chirp rate $\beta_k$ is

$$\frac{\partial \mathcal{L}}{\partial \beta} = -2x^2 \sum_k \left(\alpha_k \sin(\theta_k) - \gamma_k \cos(\theta_k)\right) \tag{12}$$

where $\theta_k = 2\pi\omega_k x + \pi\beta_k x^2$. The presence of the $x^2$ term indicates that the gradient magnitude is sensitive to $x$. The $x^2$ term increases the magnitudes $|x|$ and the Chirp rate gradients, which can accelerate the learning of edges or high-frequency noise. For inputs with small $|x|$, this term can suppress gradient noise and result in training robustness.

*2) Connection to MSRS Challenges:* The properties of ChirpFAN are directly relevant to addressing domain shifts in MSRS data. MSRS data often exhibit nonstationarities such as shifts or broadening in spectral signatures due to atmospheric effects or material variations. Similarly, spatial textures vary because of geometric distortions or changes in resolution. ChirpFAN can model varying frequencies, which allows it to adaptively capture these linearly frequency-modulated components. This property of ChirpFAN provides a more accurate representation than the fixed-frequency bases, such as FAN and wavelets. Furthermore, as Theorem 1 shows, the gradients of ChirpFAN are sensitive to input magnitude via the $x^2$ term, which benefits learning from complex MSRS data. Because edges, textural boundaries, or sharp spectral features often show larger values in processed feature maps, the amplified gradients in these regions can accelerate learning of these discriminative features.

Theoretically, the $x^2$ term could lead to gradient instability with large input values. To solve this, the standard Layer Normalization is applied to mitigate the gradient instability in our ChirpST architecture. The normalization bounds the input $x$ and ensures training stability. Then, the $x^2$ term is a data-driven and adaptive learning rate for the chirp parameter.

While the underlying physical processes may involve higher-order nonlinearities, the Chirplet transform provides a powerful first-order approximation. Distinguished from fixed-frequency or purely data-driven models, the following experimental results suggest that the Chirplet linear model is effective in capturing the main signal-level distortions.

## IV. Proposed ChirpFAN Framework

In this article, a ChirpFAN is proposed for cross-scene classification of MSRS data. Its overall flowchart is shown in Fig. 3. The proposed ChirpFAN consists of: 1) an FrSFP feature extraction part to extract multidomain features in Section IV-A and 2) a ChirpFAN-based attention module designed for multisource data analysis in Section IV-B. Then, an end-to-end network for adaptation is designed in Section IV-C. Finally, the classifier, discriminators, and the training procedures are introduced in Section IV-D.

The labeled source-domain (SD) data is denoted as $\mathcal{S} = \{(\mathcal{X}^s_{\mathcal{H}}, \mathcal{X}^s_{\mathcal{L}}), Y^s\}$, and the unlabeled target-domain (TD) data is denoted as $\mathcal{T} = \{(\mathcal{X}^t_{\mathcal{H}}, \mathcal{X}^t_{\mathcal{L}})\}$. Here, $\mathcal{X}_{\mathcal{H}} \in \mathbb{R}^{C_h \times W \times H}$ represents the HSI data and $\mathcal{X}_{\mathcal{L}} \in \mathbb{R}^{C_l \times W \times H}$ represents the LiDAR data. $H$ and $W$ are the spatial dimensions, and $C_h$ and $C_l$ are the number of channels. $Y^s$ is the source label (containing $N$ classes), while the TD label space $Y^t$ is unknown. The superscripts $s$ and $t$ denote the source and TDs, respectively.

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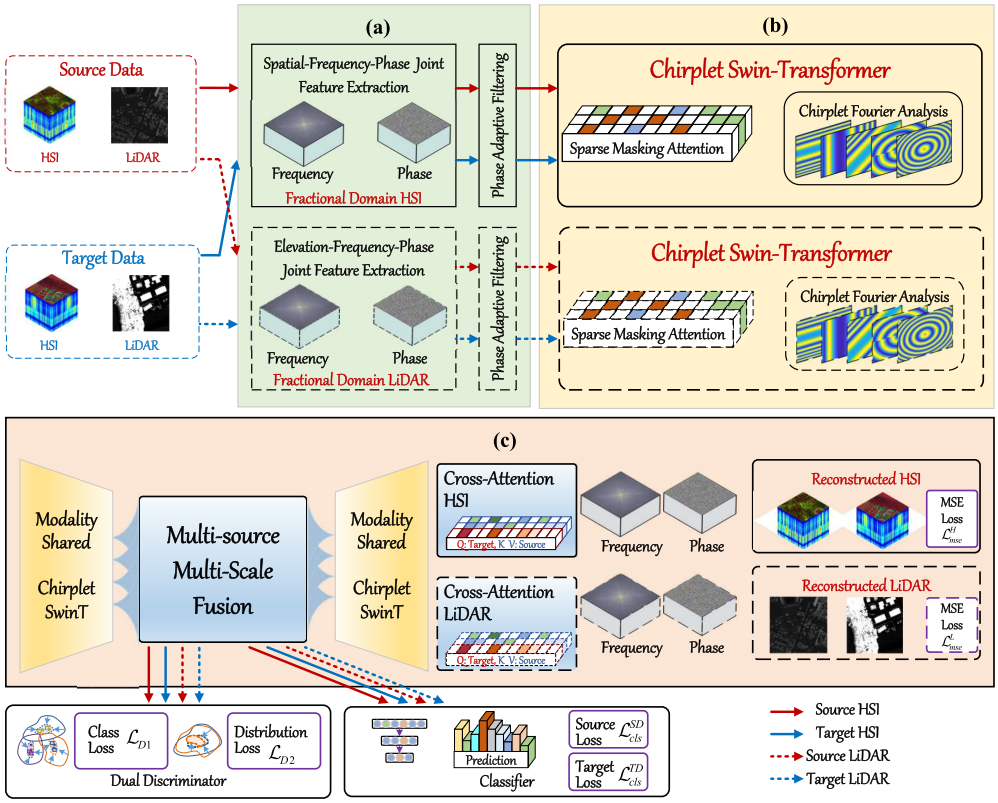



Fig. 3. Flowchart of proposed ChirpFAN framework. It contains four parts: 1) an FrSFP feature extraction part; 2) a ChirpFAN-based attention module; 3) an end-to-end network for adaptation; and 4) the classifier and discriminators. (a) Fractional feature extraction. (b) Modality-specific Chirplet analysis. (c) End-to-end modality-share network.

### A. FrSFP Feature Extraction

To address the redundancy and interference introduced during multisensor imaging, frequency-domain methods, represented by the Fourier transform and FrFT, are utilized. They can remove interference in the transform domain and capture image structures (such as edges, textures, and shapes) and variational features shown in Fig. 1. Inspired by this, we designed an FrSFP block as Fig. 3(a) shows. This block acts as a transform-domain filter and feature preextractor. By jointly using magnitude and phase, it enhances the discrimination of land covers while suppressing noise, providing augmented features for cross-domain analysis.

First, a 2-D convolution (Conv) block is used to align the channel dimensions to $C_0 \times W \times H$

$$\mathcal{X}^s_{\mathcal{H}_0} = f_{2D3}^{C_0 \to C_0}\left(f_{2D3}^{C_h \to C_0}\left(\mathcal{X}^s_{\mathcal{H}}\right)\right), \mathcal{X}^s_{\mathcal{L}_0} = f_{2D3}^{C_l \to C_0}\left(\mathcal{X}^s_{\mathcal{L}}\right) \tag{13}$$

where $f_{2D3}^{*\to C_0}(\cdot)$ represents a 2-D convolution block (Conv+BN+ReLU) with $3 \times 3$ window size, $C_0$ output channels, and a stride of 1.

Then, the FrFT is applied to the preprocessed signals $(\mathcal{X}^s_{\mathcal{H}_0}, \mathcal{X}^s_{\mathcal{L}_0})$, which transforms them into the fractional domain. For the HSI data $\mathcal{X}^s_{\mathcal{H}_0} \in \mathbb{R}^{C_0 \times H \times W}$, the complex-domain features after applying FrFT are

$$X_\alpha = \mathcal{F}_\alpha\left(\mathcal{X}^s_{\mathcal{H}_0}\right) = M_\alpha \odot e^{j\phi_\alpha} \tag{14}$$

where $\mathcal{F}_\alpha$ represents the 2-D FrFT operator of order $\alpha$, applied independently to each spatial slice $(H \times W)$ for every channel and batch element. $M_\alpha = |X_\alpha|$ is the element-wise magnitude spectrum. $\phi_\alpha = \arg(X_\alpha)$ is the phase spectrum with the principal value in $(-\pi, \pi]$. $\odot$ denotes the Hadamard product.

Then, a phase-aware weight module is designed for phase feature $\phi_\alpha \in \mathbb{R}^{C_0 \times H \times W}$, which is usually discarded in multidomain analysis. The phase spectrum reflects the structural and positional details of MSRS data. Let $f_{\text{PW1}}$ (output channels $C_0/k$) and $f_{\text{PW2}}$ (output channels $2C_0$) denote these layers

$$\begin{aligned} Z &= \text{GELU}\left(f_{\text{PW1}}\left(\phi_\alpha\right)\right) \quad \in \mathbb{R}^{(C_0/k)\times H\times W} \\ \phi_{\text{out}} &= \sigma\left(f_{\text{PW2}}\left(Z\right)\right) \quad \in \mathbb{R}^{2C_0\times H\times W}. \end{aligned} \tag{15}$$

The output $\phi_{\text{out}}$ is then split along the channel dimension into two tensors, $\phi_1$ and $\phi_2$, each of size $\mathbb{R}^{C_0 \times H \times W}$. GELU is the activation function [54].

For the magnitude spectrum, two Conv blocks $T_1 = \mathcal{C}_{5\times5}(M_\alpha)$ and $T_2 = \mathcal{C}_{7\times7}(T_1)$ are applied, where $\mathcal{C}_{k\times k}$ denotes convolution operations with $k \times k$ kernels. The multidomain phase-aware feature $\mathcal{X}^\phi_{\mathcal{H}} \in \mathbb{R}^{C_0 \times H \times W}$ of HSI is

$$\mathcal{X}^\phi_{\mathcal{H}} = \phi_1 \odot T_1 + \phi_2 \odot T_2. \tag{16}$$

Under the conditions maximizing the variance of $\mathcal{X}^\phi_{\mathcal{H}}$ componentwise, the optimal weights are related to the power of the features

$$\phi_1^* \propto \|T_1\|_F^2, \quad \phi_2^* \propto \|T_2\|_F^2 \tag{17}$$

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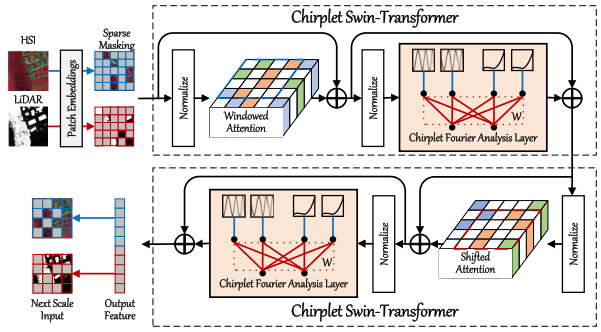



Fig. 4. Proposed ChirpST.

where $\|\cdot\|_F^2$ denotes the squared Frobenius norm (sum of squares of all elements). Then, the normalizing operation is

$$\phi_1^* = \frac{\|T_1\|_F^2}{\|T_1\|_F^2 + \|T_2\|_F^2}, \phi_2^* = \frac{\|T_2\|_F^2}{\|T_1\|_F^2 + \|T_2\|_F^2}. \tag{18}$$

Similarly, the enhanced features for the LiDAR modality, derived through the same process applied to $\mathcal{X}_{\mathcal{L}}^s$, are denoted as $\mathcal{X}_{\mathcal{L}}^{\phi} \in \mathbb{R}^{C_0 \times W \times H}$.

*Theorem 2:* If the phase weight generator subnetwork (from $\phi_\alpha$ to $\phi_1, \phi_2$) has a bounded Lipschitz constant, specifically if the operator norm of the linear layers and activations satisfy appropriate bounds [summarized as a constant $L$ for the map $\phi_\alpha \mapsto (\phi_1, \phi_2)$], then the change in the output feature $T$ is bounded with respect to changes in the input phase $\Delta\phi$

$$\|T(\phi_\alpha) - T(\phi_\alpha + \Delta\phi)\|_F \leq K\left(\|\Delta\phi\|_F\right) \tag{19}$$

where the bound $K$ depends on the Lipschitz constant $L$ of the weight generator and on bounds for the feature norms $\|T_1\|_F, \|T_2\|_F$.

This indicates that small perturbations in the phase input lead to controlled changes in the fused output feature, ensuring stability. The stability follows from the Lipschitz properties of the constituent operations and the bounded nature of the features $T_1$ and $T_2$ being modulated.

In this section, the proposed block provides: 1) an adaptive fractional order parameter $\alpha$ enabling frequency feature selection and 2) a phase-aware weighting module for optimal multidomain feature fusion.

### B. Chirplet Fourier ST

Using the multidomain amplitude-phase features extracted in Section IV-A, the enhanced features are separated into blocks via patch embedding. For each input block, a linear embedding layer projects the channel dimension from $C_0$ to $C_1$. Then, a window masking step is used in the multimodal masked learning step. With a masking ratio $\theta = 0.75$, it generates input features comprising only this subset of visible blocks, which enables the model to learn more representative multimodal features from the MSRS data. For the input HSI features $\mathcal{X}_{\mathcal{H}}^{\phi} \in \mathbb{R}^{C_0 \times W \times H}$, the unmasked blocks retained after masking are denoted as $\mathcal{M}(\mathcal{X}_{\mathcal{H}}^{\phi}) \in \mathbb{R}^{C_0 \times (1-\theta)WH}$.

With the ChirpFAN layer proposed in Section III, the ChirpST is designed to extract multimodal features from the masked features. As illustrated in Fig. 4, two Swin Transformer (ST) [55] blocks are utilized to learn intramodal amplitude and phase information.

The encoding process can be described as follows:

$$\mathcal{X}_{\mathcal{H}_1}^s = \text{ChirpST}\left(\mathcal{M}\left(\mathcal{X}_{\mathcal{H}}^{\phi}\right)\right) \tag{20}$$

where the proposed ChirpFAN layer in the ChirpST block performs time-varying frequency-domain decomposition on local window features using the Chirplet basis $\psi_{\beta,\omega}(t) = e^{j2\pi(\omega t + \beta/2t^2)}$. According to Theorem 1, ChirpFAN can approximate complex patterns with fewer basis functions

$$\left\| \mathcal{X}_{\mathcal{H}}^{\phi} - \sum_{k=1}^{M} c_k \psi_{\beta_k,\omega_k}(t) \right\|_2 < \epsilon. \tag{21}$$

Compared to the nonlinear mapping of MLPs, ChirpFAN adaptively matches the interspectral variations of HSI data. It enhances the sparsity of the feature representation through the Chirp-rate $\beta_k$. While the windowed self-attention (SA) represents local spatial features, the ChirpFAN models local nonstationary patterns and enhances local–global feature interaction. The dynamic chirp rate $\beta_k$ allows ChirpFAN to adaptively represent local frequency modulations. This adaptive filtering leads to a more compact representation where signal energy is Chirp-related. The energy concentration into fewer and more descriptive coefficients contributes to an enhanced sparsity in the feature space, which benefits the following processing. Moreover, as for the LiDAR modality, the encoded features after ChirpST can be denoted as $\mathcal{X}_{\mathcal{L}_1}^s$.

The spectra shifts of HSIs can be modeled as nonstationary signals, which can be modeled by the proposed ChirpFAN through its quadratic phase term. This outperforms FAN using fixed frequencies and MLPs employing nonlinear transforms. In classification tasks, ChirpFAN's chirplet basis functions can generalize texturally rotated and scaled patterns, while the MLP and FAN lead to overfitting because of fixed frequencies or purely data-driven transforms. Specifically, ChirpFAN provides superior modeling of nonstationary characteristics or shifts in spectral signatures for HSI classification.

In our ChirpST, the ChirpFA layer is an in-place replacement of the standard MLP layer to impose a signal-aware inductive bias. By replacing the MLP block, the network can learn the Chirplet domain representations, focusing on the signal modulations. In contrast, a parallel-block design usually weights the general-purpose MLP path, which omits the effect of Chirplet analysis in processing nonstationary spectral and spatial modulations.

### C. End-to-End Multisource Network

As shown in Fig. 3(c), a modality-shared end-to-end network is designed for the intermodal relationship. Feature consistency is enhanced through cross-modal interaction using the encoded HSI and LiDAR features in Section IV-B $\mathcal{X}_{\mathcal{H}_1}^s$ and $\mathcal{X}_{\mathcal{L}_1}^s$. The overall shared encoding process is defined as follows:

$$\left[\mathcal{X}_{\mathcal{H}_2}^s, \mathcal{X}_{\mathcal{L}_2}^s\right] = \text{SwinT}\left(\text{ChirpFAN}\left(\left[\mathcal{X}_{\mathcal{H}_1}^s, \mathcal{X}_{\mathcal{L}_1}^s\right]\right)\right). \tag{22}$$

As shown in (22), the ChirpFAN layers are applied to decompose and align the MSRS features in fractional domains.

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The cross-modal attention adjusts weights based on query–key similarities between modalities. It allows the proposed framework to learn shared representations related to frequency components and suppress intermodal interference.

In this stage, the ChirpFAN enabled the frequency-domain alignment. Through multiscale decomposition using Chirp basis functions, the spectral oscillations of HSIs and the geometric structures of LiDAR data are mapped into a unified frequency domain space. This suppresses intermodal interference. At the same time, the Chirp rates are dynamically adjusted by cross-modal attention weights, achieving frequency-adaptive fusion with the fusion weight

$$w_a = \sigma\left(\text{Softmax}\left(QK^T/\sqrt{d}\right)\right) \tag{23}$$

where $Q$ and $K$ represent the cross-modal query and key vectors, respectively, forming a frequency–space dual attention. The encoded SD multimodal and multiscale features are $\mathcal{X}^s_{\mathcal{H}_1}, \mathcal{X}^s_{\mathcal{H}_2}$ and $\mathcal{X}^s_{\mathcal{L}_1}, \mathcal{X}^s_{\mathcal{L}_2}$. The TD features are denoted as $\mathcal{X}^t_{\mathcal{H}_1}, \mathcal{X}^t_{\mathcal{H}_2}$ and $\mathcal{X}^t_{\mathcal{L}_1}, \mathcal{X}^t_{\mathcal{L}_2}$.

As shown in Fig. 3(c), the decoder comprises a shared decoder and two modality-specific decoders for cross-modal reconstruction. The shared decoder $f_{\text{sha-de}}(\cdot)$ uses two ChirpST blocks and two upsampling operations to restore spatial resolution as follows:

$$\mathcal{X}^{s'}_{\mathcal{H}_1} = f_{\text{sha-de}}\left(\mathcal{X}^s_{\mathcal{H}_2}\right), \quad \mathcal{X}^{s'}_{\mathcal{L}_1} = f_{\text{sha-de}}\left(\mathcal{X}^s_{\mathcal{L}_2}\right) \tag{24}$$

where $\mathcal{X}^s_{\mathcal{H}_2} \in \mathbb{R}^{C_3 \times W/4 \times H/4}$ is the encoder output.

Then, modality-specific decoders are designed to utilize multimodal complementarity for cross-modal reconstruction. Taking the reconstruction of HSI multidomain features as an example, it contains SA and cross-attention (CA) interaction. For SA, the reconstructed HSI modal features $\mathcal{X}^{s'}_{\mathcal{H}_1}$ are projected into query $q_h$, key $k_h$, and value $v_h$ vectors. This module learns intramodal frequency domain priors and obtains the SA map $A_{\text{sa}}$, which enhances the capability to model spectral continuity. For CA, the concatenated features $[\mathcal{X}^{s'}_{\mathcal{H}_1}, \mathcal{X}^{s'}_{\mathcal{L}_1}]$ are mapped to form intermodal keys to integrate multimodal complementarity for precise HSI reconstruction. The geometric features from LiDAR are mapped via frequency modulation projection to form cross-modal keys $k_l$. The frequency domain matching degree between these keys and the HSI query $q_h$ determines the cross-modal fusion attention $A_{\text{ca}}$.

By fusing $A_{\text{sa}}$ and $A_{\text{ca}}$ and then multiplying them with the value $v_h$, the HSI modality $\mathcal{X}^{s'}_{\mathcal{H}_1}$ is reconstructed as follows:

$$\mathcal{X}^{s'}_{\mathcal{H}_1} = f_o\left((A_{\text{sa}} \oplus A_{\text{ca}}) \otimes v_h\right). \tag{25}$$

Then, using the multidomain features, the reconstructed data $\mathcal{X}^{s'}_{\mathcal{H}_0}$ can be represented as follows:

$$\mathcal{X}^{s'}_{\mathcal{H}_0} = \mathcal{F}^{-1}_{\alpha}\left(\sum_{\beta} w_\beta \cdot \mathcal{F}_{\alpha}\left(\mathcal{X}^{s'}_{\mathcal{H}_1}\right) \odot \mathcal{F}^{-1}_{\alpha}\left(\mathcal{X}^{v'}_{\mathcal{L}_1}\right)\right) \tag{26}$$

where $w_\beta$ are generated by the frequency weighting layer and implement frequency-domain cross-modulation, $\mathcal{X}^{v'}_{\mathcal{L}_1}$ is the LiDAR value. In the decoder, the ChirpST blocks reconstruct detailed features while suppressing aliasing noise. The frequency modulation projection used for LiDAR keys $k_l$ enables matching with HSI queries $q_h$. Within a domain sensitive to textural details encoded by frequency variations, the cross-model attention improves the reconstructed $\mathcal{X}^{s'}_{\mathcal{H}_0}$. Similarly, the reconstructed LiDAR data can be denoted as $\mathcal{X}^{s'}_{\mathcal{L}_0}$.

Based on the above design, the time–frequency localization property enables the proposed ChirpFAN to suppress aliasing noise, and the reconstruction error satisfies

$$\left\|\mathcal{X}^{s'}_{\mathcal{H}_0} - \mathcal{X}^{s}_{\mathcal{H}_0}\right\|_2 \le \lambda \cdot \text{TV}\left(\mathcal{X}^{s}_{\mathcal{H}_0}\right) \tag{27}$$

where TV represents the total variation regularizer and $\lambda$ is controlled by the Chirp rates.

Through this multimodal mask learning, multimodal features and their reconstructions can be extracted using only a small part of the visible patches. Introducing ChirpFAN into the masked autoencoder architecture mathematically optimizes the multimodal features representation. Furthermore, it enhances the reconstruction accuracy and generalization of the proposed network for cross-scene classification of MSRS data.

### D. Network Training

In this section, we design the network with the encoded SD multimodal multiscale features $\mathcal{F}^s = [\mathcal{X}^s_{\mathcal{H}_1}, \mathcal{X}^s_{\mathcal{L}_1}, \mathcal{X}^s_{\mathcal{H}_2}, \mathcal{X}^s_{\mathcal{L}_2}]$ and the TD features $\mathcal{F}^t = [\mathcal{X}^t_{\mathcal{H}_1}, \mathcal{X}^t_{\mathcal{L}_1}, \mathcal{X}^t_{\mathcal{H}_2}, \mathcal{X}^t_{\mathcal{L}_2}]$.

A multimodal feature fusion and classifier module is constructed to map the features to class using deconvolutional and convolutional layers and then predict the final SD classification map $Y^s_p$ as follows:

$$Y^s_p = C\left(\mathcal{F}^s\right) = f^{128\to N}\left(f^{6C_f\to 128}\left(\mathcal{F}^s\right)\right) \tag{28}$$

where $f^{6C_0\to 128}(\cdot)$ is a 2-D deconvolutional layer with output channels 128 and stride 2, and $f^{128\to N}(\cdot)$ is a convolutional layer, with the output $N$ being the number of land covers. Therefore, the classifier loss is $\mathcal{L}_C = \mathcal{L}_C(Y^s, Y^s_p)$.

Then, a domain adaptation module based on dual discriminators is designed, focusing on the domain shifts. As shown in Fig. 3, this module includes a local fine-grained class discriminator $D_1$ and an entropy-constrained global discriminator $D_2$.

$D_1$ focuses on class variations, such as the tree class that varies across seasons. Instead of traditional 0 and 1 domain encoding, multichannel soft labels are used in $D_1$. The SD domain encoding is $[0; E^s]$ and the TD domain encoding is $[E^t; 0]$. Then, the SD per-channel calculation is $E_n = (e^{(Y_j)^{N/T}})/\left(\sum_{j=1}^{N} e^{(Y_j)/T}\right)$, where the coefficient $T = 1.8$ is used to smooth the probability distribution. $E_n$ is then thresholded with confidence threshold $\theta = 0.9$. Thus, the discriminator $D_1$ loss $\mathcal{L}_{\text{dis1}}$ and adversarial loss $\mathcal{L}_{\text{adv1}}$ are defined as follows:

$$\mathcal{L}_{\text{dis1}} = -\left[E^s \log D_1\left(F^s\right) + E^t \log\left(1 - D_1\left(F^t\right)\right)\right] \tag{29}$$

$$\mathcal{L}_{\text{adv1}} = -E^t \log\left(D_1\left(F^t\right), 0\right). \tag{30}$$

The global entropy-constrained discriminator $D_2$ is used to prevent model overfitting and improve generalization. The target is to make the entropy map of SD and TD similar, achieving cross-domain global structure consistency. Shannon entropy $H(\cdot)$ is calculated based on pixel-level prediction results $(Y^s_p, Y^t_p)$ and then an adversarial learning strategy is used for entropy-level domain adaptation. The discriminator loss of $D_2$ is

$$\mathcal{L}_{\text{dis2}} = \mathcal{L}_{\text{bce}}\left(D_2\left(Y^s_p\right), 0\right) + \mathcal{L}_{\text{bce}}\left(D_2\left(Y^t_p\right), 1\right) \tag{31}$$

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**Algorithm 1** ChirpFAN Classification Framework

**Input:** Source data $(\mathcal{X}_{\mathcal{H}}^{s}, \mathcal{X}_{\mathcal{L}}^{s})$, source labels $\mathbf{Y}^{s}$, target data $(\mathcal{X}_{\mathcal{H}}^{t}, \mathcal{X}_{\mathcal{L}}^{t})$.
**Output:** Best trained models $G, D_1, D_2$, Target labels $\mathbf{Y}^{T}$.
**procedure** TRAINING($args.epoch$)
  Data patch preparation and initialize all weights and bias terms.
  Initialize Generator $G$ as Sec. IV-A–C and Discriminators $D_1, D_2$ as Sec. IV-D.
  **for** $ep = 0$**to**$args.epoch - 1$ **do**
    Train Generator G Using SD data: $\mathcal{F}^{s}, \mathbf{P}^{s}, \mathcal{L}_{\text{mse}}^{s} \leftarrow G(\mathbf{X}^{s}, \text{'source'})$
    Train Generator G Using TD data: $\mathcal{F}^{t}, \mathbf{P}^{t}, _ \leftarrow G(\mathbf{X}^{t}, \text{'target'})$
    Train Discriminators $D_1, D_2$.
  **end for**
**end procedure**
**procedure** TEST($G, D_1, D_2$)
  $\mathbf{Y}_{\text{pred_list}} \leftarrow [\,]$, $\mathbf{Y}_{\text{gt_list}} \leftarrow [\,]$
  **for all** $(\mathbf{X}_{\text{b}}, \mathbf{Y}_{\text{gt_b}})$ in Dataloader **do**
    $_, \mathbf{P}_{\text{b}}, _ \leftarrow G(\mathbf{X}_{\text{b}}, [D_1, D_2], \text{'target'})$
    $\mathbf{Y}_{\text{pred_b}} \leftarrow \text{argmax}(\mathbf{P}_{\text{b}})$
    Append $\mathbf{Y}_{\text{pred_b}}$ to $\mathbf{Y}_{\text{pred_list}}$; $\mathbf{Y}_{\text{gt_b}}$ to $\mathbf{Y}_{\text{gt_list}}$
  **end for**
  Update the predictions $\mathbf{Y}^{T}, \mathbf{Y}_{\text{gt}}$
  **return** $\mathbf{Y}^{T}$
**end procedure**

and the adversarial loss is

$$\mathcal{L}_{\text{adv2}} = \mathcal{L}_{\text{bce}}\left(D_2\left(H\left(Y_p^{t}\right)\right), 0\right) \tag{32}$$

where $\mathcal{L}_{\text{bce}}(\cdot)$ being the binary cross-entropy loss function.

Considering the dual discriminators, the total discriminator loss $\mathcal{L}_{\text{dis}}$ is

$$\mathcal{L}_{\text{dis}} = \mathcal{L}_{\text{dis1}} + \lambda_1 \mathcal{L}_{\text{dis2}} \tag{33}$$

and total adversarial loss $\mathcal{L}_{\text{adv}}$ is

$$\mathcal{L}_{\text{adv}} = \mathcal{L}_{\text{adv1}} + \lambda_2 \mathcal{L}_{\text{adv2}} \tag{34}$$

where the weighting factors are $\lambda_1 = \lambda_2 = 0.1$, which achieves finer-grained domain alignment and improves cross-domain classification accuracy.

The overall discriminator objective of proposed network is captured by $\mathcal{L}_D = \mathcal{L}_{\text{dis}} - \mathcal{L}_{\text{adv}}$. The data reconstruction loss is defined as $\mathcal{L}_G = \mathcal{L}_{\text{mse}}(\mathcal{X}_{\mathcal{H}}) + \mathcal{L}_{\text{mse}}(\mathcal{X}_{\mathcal{L}})$. The overall objective function is

$$\arg\min_{G,C} \arg\max_{D} \left(\mathcal{L}_G + \mathcal{L}_C - \mathcal{L}_D\right). \tag{35}$$

During the entire end-to-end training process, the 75% sparse masking is applied. This strategy is a regularizer to make the discriminators ($D_1$ and $D_2$) and the classifier ($C$) operate on a sparse subset of visible patches. Then, the network is prevented from overfitting and can learn robust and context-aware representations, which improve the generalization capability for the unseen TD. The detailed training and inference procedures of the proposed ChirpFAN are summarized in Algorithm 1.

## V. Experimental Results and Comparison

### A. Experimental Data

In this section, three cross-domain multimodal remote sensing (MSRS) datasets are utilized to evaluate the effectiveness of the proposed ChirpFAN.

*1) Houston13–Houston18 (H13–H18) Scenes:* The dataset consists of HSI and LiDAR data acquired over the University of Houston campus (USA) in 2013[1] and 2018,[2] respectively. For cross-temporal analysis, the overlapping area was processed, involving resolution matching (downsampling Houston18) and the extraction of 48 common HSI spectral bands across seven shared land-cover classes.

*2) Nashua–Hanover (Na–Ha) Scenes:* Sourced from G-LiHT data,[3] the dataset includes co-registered HSI and LiDAR data from Nashua (450 × 1050 pixels) and Hanover (700 × 620 pixels), USA, both at 1-m ground sample distance (GSD). The HSIs feature 114 spectral bands (0.42–0.95 $\mu$m). There are seven land-cover classes used for cross-city analysis.

*3) Houston–Trento (H13–T) Scenes:* The dataset combines the Houston13 HSI/LiDAR data with HSI/LiDAR data acquired over a rural area in Trento, Italy [56]. For cross-city analysis, 64 common HSI spectral bands were extracted across four shared land-cover classes.

### B. Experimental Setup

The experimental programs are implemented using Python 3.12.0 and Pytorch 2.6.0 on a personal computer equipped with NVIDIA GeForce RTX 4060 Ti.

*1) Algorithm Parameter Configuration:* The proposed ChirpFAN uses two sets of parameters. The Adam optimizer is used in the training phase. No regularization term has been applied to the cost functions. The learning rate can be updated by multiplying the initial learning rate by 1−(epoch)/(Epochs), where the weight-decay is $10^{-3}$, and the basic learning rate lr = $5 \times 10^{-4}$ will be researched in Section V-C. The input batch size is 4, with the patch size being 64. In the fractional feature extraction part, $(2\alpha)/\pi = 0.4$ order FrFT is applied. In the ChirpST module, each attention block contains a $\theta = 75\%$ masked attention layer, a layer normalization, and a linear rate $g = 0.4$ of the ChirpFAN activation layer. For the domain discriminator, the learning rate is $\text{lr}_D = 10^{-4}$. The overall network consists of four modality-specific ChirpST blocks and four modality-shared ChirpST blocks. Each ChirpST block contains eight attention heads.

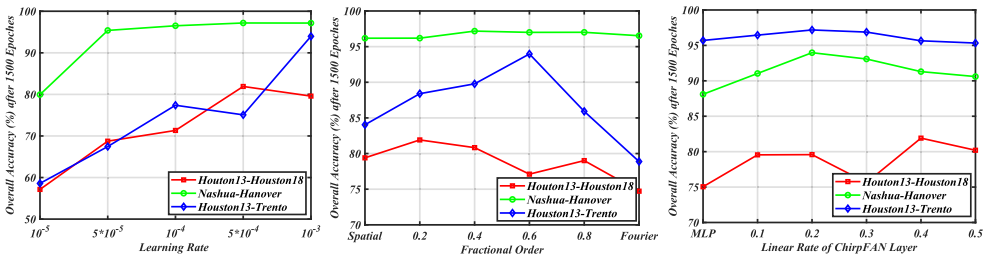
*2) Performance Analysis Configuration:* To validate the proposed ChirpFAN's effectiveness quantitatively and qualitatively, classification performance on three cross-scene datasets is compared with that of other competitive classifiers. The compared state-of-the-art (SOTA) methods include fractional fusion and spatial–spectral-domain adaptation (FrF-SSDA) [25], supervised contrastive learning-based unsupervised domain adaptation (SCLUDA) [32], high-resolution domain adaptation networks (HighDANs) [12], masked self-distillation

[1] http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

[2] http://www.grss-ieee.org/community/technical-committees/data-fusion/2018-ieee-grss-data-fusion-contest/

[3] https://glihtdata.gsfc.nasa.gov

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(a) (b) (c)

Fig. 5. Classification performance of the proposed ChirpFAN with different parameters. (a) Learning rate lr. (b) Fractional order $(2\alpha)/\pi$. (c) Linear rate of the Chirplet layer.

domain adaptation (MSDA) [33], multilevel unsupervised domain adaptation (MLUDA) [31], and shared–private feature alignment semi-supervised (SASS) learning [28], as well as two time–frequency analysis methods: Wavelet-CNN [46] and Gabor-based CNN [48]. The experimental setups of the compared methods are optimized as suggested. All compared methods use original image patches as inputs without data augmentation. Three commonly used evaluation metrics are adopted, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa), to quantify the experimental results. To assess statistical significance, ten experiments are conducted, and the standard error is computed.

### C. Parameter Analysis

To validate the effectiveness and sensitivity of parameters involved in the proposed ChirpFAN, experimental analysis using varying parameters is compared in Fig. 5.

*1) Learning Rate lr:* In Fig. 5(a), the OA after 1500 training epochs shows that the learning rate lr affects the convergence of the learning process. The search range is constrained from $10^{-5}$ to $10^{-3}$. The algorithm using a small learning rate, such as $10^{-5}$, cannot converge effectively and results in poor performance, with OA below 70% for all three datasets. The OA improves as the learning rate increases, but large learning rates like $10^{-3}$ can lead to large fluctuations in the objective function and result in an unstable training process. The optimal learning rates appear to be around $10^{-4}$ to $5 \times 10^{-4}$, where most datasets achieve their peak performance.

*2) Fractional Order $(2\alpha)/\pi$:* Fig. 5(b) illustrates the impact of the fractional order $(2\alpha)/\pi$ on the OA after 1500 epochs. The fractional order $(2\alpha)/\pi$ is varied from spatial domain $(2\alpha)/\pi = 0$ to the Fourier domain $(2\alpha)/\pi = 1$. For the scene sensitive to local spatial feature, for example, the H13–H18 dataset, the performance is better when using a small $(2\alpha)/\pi$ with more spatial features in mixed features. The Na–Ha dataset shows a peak performance around $(2\alpha)/\pi = 0.4$. For the H13–T dataset focusing on global features, the OA shows a peak at $(2\alpha)/\pi = 0.6$ and decreases as $(2\alpha)/\pi$ approaches the Fourier domain. Generally, larger fractional orders indicate more global frequency features, while the optimal $(2\alpha)/\pi$ is decided by the resolutions of datasets. In the following research, a learnable fractional order $\alpha$ in the FrFT module would enable the feature extraction module to find the optimal analysis domain as (4) instead of relying on a fixed value.

TABLE I
STATISTICAL ANALYSIS AND PHYSICAL PROPERTIES OF LEARNED $\beta_k$ DISTRIBUTIONS

| Feature | HSI Branch | LiDAR Branch |
|---|---|---|
| **Learned Chirp** | **Up-Chirp** | **Down-Chirp** |
| **Mean ($\mu$)** | 0.182 | $-0.179$ |
| **Std. Dev. ($\sigma$)** | 0.051 | 0.063 |
| **Range [Min, Max]** | $[0.02, 0.40]$ | $[-0.25, -0.02]$ |

*3) Linear Rate of the Chirplet Layer $g$:* Fig. 5(c) shows the effect of the linear rate $g$ of the ChirpFAN layer. Larger $g$ means more linear features are kept, while $g = 0$ means pure nonlinear layer, and $g = 0.5$ means pure linear layer. The parameter $g$ is varied from $g = 0$ to 0.5. For the H13–H18 dataset, the OA shows a peak at $g = 0.4$. The Na–Ha and H13–T datasets show the highest performance at $g = 0.2$, decreasing as $g$ moves toward 0.5.

*4) Analysis of Learned Chirplet Rate $\beta_k$:* To analyze the learned Chirplet rate $\beta_k$ and validate its learning stability, we examined its distributions from the $W_\beta$ weights after training. Table I shows these distributions in both the HSI and LiDAR modality-specific branches. The distributions of learned $\beta_k$ parameters are stable and not collapsing to zero. Thus, the network is actively using the chirp rate to model nonstationary, frequency-modulated components. Combined with the gradient stability provided by the Layer Normalization, the proposed ChirpFA demonstrates stable learning behavior.

As Table I shows, the model learns different chirp rates for HSI and LiDAR, which suggests it learns signal-level modulations of different sensors. For the spectral chirps in HSIs, the learned $\beta_k$ is a positive up-chirp showing the spectral shifts. These spectral domain shifts are caused by atmospheric effects or illumination changes, which are direct forms of spectral frequency modulation. For the spatial chirps in LiDAR, the negative-biased $\beta_k$ shows that the model is capturing spatial chirps related to geometric and perspective distortions. For example, as the sensor's view angle changes between scenes, spatial textures appear compressed with high spatial frequency at one end and stretched with low spatial frequency at the other, which is represented as a classic down-chirp. The model learns specific $\beta_k$ distributions for the spectral chirps in HSI and the spatial chirps in LiDAR, which can adapt to specific physical, signal-level modulations of sensors.

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TABLE II
ABLATION STUDY OF CHIRPFAN ON THE THREE CROSS-SCENE MSRS DATASETS

| | Modules | | | | | | | | | H13-18 | | Na-Ha | | H13-T | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Spatial | Frequency | Phase | FrFT | MLP | FAN | GaborST | WaveletST | ChirpFAN | OA | Kappa | OA | Kappa | OA | Kappa |
| I | √ | × | × | × | × | × | × | × | √ | 76.40 | 64.84 | 90.18 | 90.12 | 84.04 | 71.33 |
| | × | √ | × | × | × | × | × | × | √ | 74.72 | 60.57 | 92.53 | 91.11 | 78.88 | 62.74 |
| | √ | √ | √ | × | × | × | × | × | √ | 78.83 | 66.6 | 93.81 | 91.01 | 89.42 | 79.82 |
| II | √ | √ | √ | √ | √ | × | × | × | × | 75.05 | 61.47 | 91.41 | 88.02 | 88.10 | 77.98 |
| | √ | √ | √ | √ | × | √ | × | × | × | 79.71 | 67.45 | 94.87 | 92.76 | 91.41 | 83.45 |
| | √ | √ | √ | √ | × | × | √ | × | × | 80.56 | 69.12 | 95.14 | 93.15 | 91.87 | 84.21 |
| | √ | √ | √ | √ | × | × | × | √ | × | 80.14 | 68.53 | 95.07 | 92.95 | 91.53 | 83.74 |
| | √ | √ | √ | √* | × | × | × | × | √ | 79.25 | 67.15 | 94.56 | 92.41 | 91.37 | 83.63 |
| | √ | √ | √ | √ | × | × | × | × | √ | **81.91** | **71.23** | **97.16** | **96.01** | **93.97** | **87.75** |

*This row was conducted using the Simple Fusion: a concatenation of magnitude and phase.

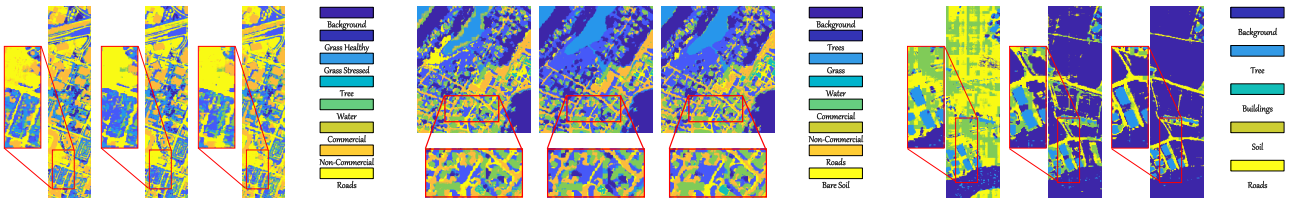



Fig. 6. Ablation analysis based on classification maps of different analysis layers. (a) MLP on the H13–H18 dataset. (b) FAN on the H13–H18 dataset. (c) ChirpFAN on the H13–H18 dataset. (d) Legend of the H13–H18 dataset. (e) MLP on the Na–Ha dataset. (f) FAN on the Na–Ha dataset. (g) ChirpFAN on the Na–Ha dataset. (h) Legend of the Na–Ha dataset. (i) MLP on the H13–T dataset. (j) FAN on the H13–T dataset. (k) ChirpFAN on the H13–T dataset. (l) Legend of the H13–T dataset.

### D. Ablation Analysis

To measure the contributions of the FrFT feature extractor and ChirpFAN modules to the performance of the proposed ChirpFAN, a detailed ablation study is conducted. The baseline models use only spatial–spectral features and use MLP layers for analysis. The proposed ChirpFAN with and without each component is denoted as (√) and (×). Table II reports the experimental results for the three cross-scene MSRS datasets. For the sake of comparative fairness, no pretrained models or processes are provided for all experiments.

*1) Analysis of the Fractional Feature Extraction:* In Part I of Table II, the feature extractor's effectiveness is enhanced when jointly using spatial, frequency, and phase features as input to the ChirpST. The combination yields the highest OA and Kappa scores across all datasets. Among these features, spatial features describe the basic geometric and structural properties, while frequency features and phase features align the spatial patterns. By integrating these varied perspectives, the ChirpFAN module receives a discriminative feature set and improves the classification results. To evaluate the effect of the phase-aware weighting module, a simple fusion variant is tested by replacing the module with a simple concatenation of magnitude and phase, followed by a $1 \times 1$ convolutional layer for fusion. As shown in Table II, the concatenation resulted in a 2%–3% drop in OA, which further proves the effectiveness of the proposed phase-aware weighting module.

*2) Analysis of the ChirpFAN Layer:* As listed in Part II of Table II, the proposed ChirpFAN analysis module is compared with traditional MLP and recently developed FAN. With the joint extraction of spatial–frequency–phase features, simply analyzing the nonlinear spatial features or linear frequency features is highly susceptible to redundant and incomplete information stacking. Thus, the proposed ChirpFAN can analyze the extracted results in Part I and consistently yields the higher OA and Kappa scores than the other two single-domain methods across all three datasets. As shown in Fig. 6, we further compare MLP, FAN, and ChirpFAN by visualizing the classification maps. As expected, both the FAN and ChirpFAN combine frequency and phase features and obtain clear boundaries. Particularly, the proposed ChirpFAN layer can better represent the sharp edge region of patches, which improves the feature discrimination of mixed land covers. Compared with the baseline models, the proposed ChirpFAN network fuses linear transform and nonlinear learning contexts to acquire both global and local dependencies. Compared with the single domain models, the proposed ChirpFAN model gains 6%–10% improvements quantitatively.

To further validate the effectiveness of the proposed adaptive Chirplet layer, WaveletST and GaborST are used to replace the ChirpFA layer with 2-D Haar Wavelets and 2-D Gabor filters. As shown in Table II, WaveletST and GaborST outperform the standard MLP and FAN, but are outperformed by the proposed ChirpFAN. The learnable chirp rate enables the proposed method to model the complex frequency modulations of cross-scene domain shifts.

### E. Classification Performance Analysis

To evaluate the effectiveness of the proposed ChirpFAN, the classification performance of the compared SOTA methods is shown in this section. The number of training samples are detailed in Tables III–V. The same training and testing samples are used for a fair comparison.

*1) Performance on the H13–H18 Scenes:* The qualitative results of competitive methods and the proposed ChirpFAN on the H13–H18 dataset are shown in Fig. 7(f)–(l), with corresponding accuracies presented in Table III. From the

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TABLE III

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE H13–H18 SCENES

| Class | Training/All Houston 13 | Test Houston 18 | FrF-SSDA 2022 [25] | SCLUDA 2023 [32] | HighDAN 2023 [12] | MSDA 2024 [33] | MLUDA 2024 [31] | SASS 2024 [28] | Gabor-CNN 2021 [48] | Wavelet-CNN 2018 [46] | ChirpFAN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Grass Healthy | 50/345 | 1353 | 53.22±1.54 | 87.65±0.88 | 81.01±1.12 | 4.29±0.50 | 65.27±2.10 | 73.45±1.30 | 70.15±1.61 | 72.33±1.45 | 76.94±0.95 |
| 2. Grass Stressed | 50/365 | 4888 | 3.17±0.75 | 69.67±1.95 | 74.30±1.60 | 88.46±0.70 | 72.62±1.88 | 64.28±2.20 | 60.22±2.11 | 65.18±1.96 | 80.63±1.15 |
| 3. Trees | 50/365 | 2007 | 28.85±3.10 | 78.18±1.35 | 91.13±0.65 | 81.27±1.25 | 59.31±3.50 | 89.91±0.80 | 65.13±3.04 | 71.06±2.53 | 93.87±0.45 |
| 4. Water | 50/285 | 22 | 86.36±2.50 | 45.45±4.10 | 90.91±1.00 | 100.00±0.00 | 90.91±1.50 | 97.11±0.60 | 91.05±1.82 | 91.53±1.51 | 95.45±0.85 |
| 5. Commercial | 50/319 | 5347 | 24.61±2.80 | 74.68±1.40 | 69.74±1.75 | 82.12±1.10 | 78.74±1.30 | 100.00±0.00 | 68.48±2.21 | 70.23±1.91 | 77.18±1.20 |
| 6. Non-commercial | 50/408 | 32459 | 95.19±0.60 | 71.19±1.80 | 77.43±1.30 | 80.09±1.05 | 71.81±1.90 | 91.00±0.75 | 75.17±1.10 | 78.68±0.93 | 84.42±0.90 |
| 7. Roads | 50/443 | 6565 | 24.74±3.20 | 53.94±2.50 | 66.00±2.10 | 67.57±1.90 | 67.41±2.30 | 66.10±2.00 | 55.37±2.52 | 61.52±2.14 | 71.28±1.50 |
| OA | | | 66.69 ± 1.10 | 69.96 ± 1.25 | 75.59 ± 0.90 | 77.66 ± 0.85 | 71.38 ± 1.40 | 78.20 ± 0.80 | 70.40 ± 1.30 | 73.15 ± 1.10 | **81.91 ± 0.65** |
| AA | | | 45.16 ± 1.80 | 68.68 ± 1.50 | 78.65 ± 1.15 | 71.97 ± 1.00 | 72.30 ± 1.70 | **83.80 ± 0.95** | 69.37 ± 1.70 | 72.93 ± 1.50 | 82.83 ± 0.70 |
| Kappa×100 | | | 36.93 ± 2.00 | 55.45 ± 1.75 | 62.73 ± 1.35 | 64.95 ± 1.20 | 56.78 ± 1.95 | 61.62 ± 1.55 | 55.10 ± 2.10 | 58.20 ± 1.80 | **71.23 ± 0.85** |

TABLE IV

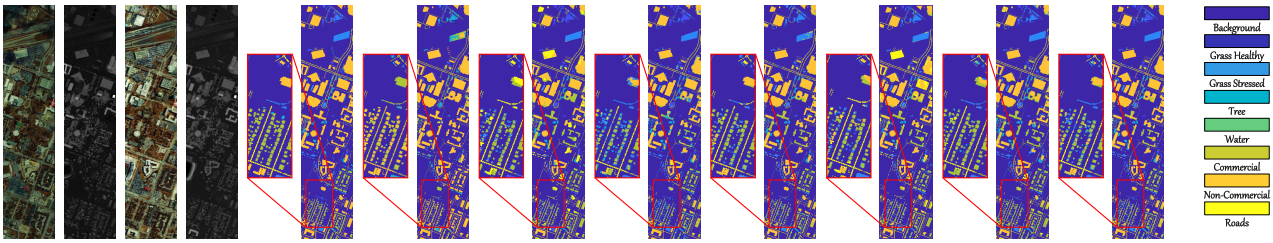
COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE NA–HA SCENES

| Class | Training/All Nashua | Test Hanover | FrF-SSDA 2022 [25] | SCLUDA 2023 [32] | HighDAN 2023 [12] | MSDA 2024 [33] | MLUDA 2024 [31] | SASS 2024 [28] | Gabor-CNN 2021 [48] | Wavelet-CNN 2018 [46] | ChirpFAN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Trees | 50/12713 | 18605 | 81.92±1.35 | 88.73±0.90 | 96.09±0.50 | 93.27±0.75 | 85.05±1.50 | 91.69±0.80 | 89.13±1.12 | 90.51±0.95 | 98.50±0.30 |
| 2. Grass | 50/7528 | 4410 | 99.18±0.40 | 100.00±0.00 | 55.96±3.50 | 99.95±0.05 | 97.98±0.60 | 100.00±0.00 | 99.16±0.32 | 99.52±0.36 | 100.00±0.00 |
| 3. Water | 50/16523 | 5425 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 |
| 4. Commercial | 50/698 | 860 | 63.84±4.10 | 100.00±0.00 | 100.00±0.00 | 100.00±0.00 | 99.98±0.02 | 100.00±0.00 | 98.35±0.42 | 99.19±0.27 | 100.00±0.00 |
| 5. Non-commercial | 50/3417 | 4338 | 93.64±0.85 | 92.52±1.10 | 93.71±0.95 | 97.65±0.45 | 98.35±0.40 | 89.46±1.25 | 90.75±1.04 | 91.84±0.98 | 93.15±0.70 |
| 6. Roads | 50/17476 | 4955 | 66.52±3.20 | 92.04±1.30 | 91.52±1.40 | 88.27±1.60 | 92.33±1.20 | 97.68±0.65 | 85.67±1.52 | 88.74±1.25 | 92.11±1.00 |
| 7. Bare Soil | 50/0821 | 764 | 0.00±0.00 | 97.11±0.70 | 69.50±2.80 | 71.47±2.50 | 95.44±0.90 | 92.65±1.15 | 71.06±2.83 | 75.01±1.86 | 83.77±1.80 |
| OA | | | 83.31±1.05 | 92.48±0.80 | 90.65±0.95 | 94.39±0.60 | 90.24±1.10 | 94.28±0.70 | 90.15 ± 0.90 | 91.82 ± 0.85 | **97.16±0.40** |
| AA | | | 72.13±1.90 | 95.77±0.55 | 86.68±1.25 | 92.94±0.85 | 95.30±0.75 | **95.93±0.65** | 90.58 ± 1.14 | 92.11 ± 0.96 | 95.36±0.50 |
| Kappa×100 | | | 77.38±1.40 | 89.87±0.90 | 87.03±1.15 | 92.23±0.75 | 86.99±1.30 | 92.18±0.85 | 87.04 ± 1.20 | 89.13 ± 1.23 | **96.01±0.45** |

TABLE V

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE H13–T SCENES

| Class | Training/All Houston 13 | Test Trento | FrF-SSDA 2022 [25] | SCLUDA 2023 [32] | HighDAN 2023 [12] | MSDA 2024 [33] | MLUDA 2024 [31] | SASS 2024 [28] | Gabor-CNN 2021 [48] | Wavelet-CNN 2018 [46] | ChirpFAN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Trees | 50/365 | 9123 | 68.71±2.15 | 85.85±1.20 | 82.10±1.60 | 93.02±0.80 | 87.75±1.40 | 93.13±0.75 | 88.17±1.34 | 90.22±1.15 | 99.33±0.25 |
| 2. Buildings | 50/319 | 2903 | 99.52±0.30 | 75.94±2.50 | 92.35±1.00 | 53.74±4.50 | 15.28±5.00 | 3.96±2.00 | 71.04±2.83 | 75.47±2.29 | 86.50±1.80 |
| 3. Ground | 50/650 | 479 | 95.62±0.90 | 1.25±1.00 | 82.46±1.90 | 71.61±3.10 | 59.82±3.80 | 74.74±2.90 | 61.28±3.14 | 65.19±2.94 | 0.00±0.00 |
| 4. Roads | 50/443 | 3174 | 72.02±2.80 | 15.41±3.50 | 55.86±4.00 | 79.46±2.20 | 70.97±3.00 | 78.85±2.40 | 71.31±2.52 | 72.16±2.24 | 92.75±1.10 |
| OA | | | 74.43±1.70 | 82.89±1.50 | 79.39±1.85 | 84.53±1.30 | 73.09±2.60 | 89.97±1.00 | 81.51 ± 1.94 | 84.22 ± 1.76 | **93.97±0.60** |
| AA | | | **83.97±1.10** | 44.61±2.80 | 78.19±2.05 | 74.45±2.30 | 58.45±3.15 | 62.67±2.55 | 72.95 ± 2.42 | 75.76 ± 2.16 | 69.65±1.30 |
| Kappa×100 | | | 60.16±2.30 | 30.15±3.90 | 63.90±2.70 | 68.48±2.90 | 54.02±4.10 | 34.11±3.30 | 65.46 ± 2.82 | 68.85 ± 2.57 | **87.75±1.50** |




(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 7. H13–H18 cross-domain classification maps. (a) Source HSI (Rband:25, Gband:14, Bband:6). (b) Source LiDAR. (c) Target HSI (Rband:25, Gband:14, Bband:6). (d) Target LiDAR. (e) Ground truth. (f) FrF-SSDA [25] (66.69%). (g) SCLUDA [32] (69.96%). (h) HighDAN [12] (75.59%). (i) MSDA [33] (77.66%). (j) MLUDA [31] (71.38%). (k) SASS [28] (78.20%). (l) ChirpFAN (81.91%).

classification maps, methods like FrF-SSDA with an OA of 66.69% show misclassification in distinguishing grass healthy, grass stressed, and tree. Semi-supervised domain adaptation methods like SCLUDA (69.96% OA) and MLUDA (71.38% OA) also rely on training sample numbers and exhibit visual misclassifications. For instance, from the SCLUDA map in Fig. 7(g), there are noticeable areas where Roads are incorrectly classified. Based on fractional multidomain feature fusion and alignment, the proposed ChirpFAN can align source and TDs with joint spectral-spatial–frequency–phase features. By reconstructing multiscale multidomain features, the fused Chirplet features can enhance the classification of textural details and targets affected by environmental variations. The proposed ChirpFAN in Fig. 7(l), with an OA of 81.91%, shows better delineation between different land-cover types, especially in complex areas with mixed classes. For instance, the distinction between commercial and noncommercial areas, and the accurate mapping of roads. Quantitatively, as shown in Table III, the proposed ChirpFAN achieves the highest OA, outperforming all listed competitive methods on the H13–H18

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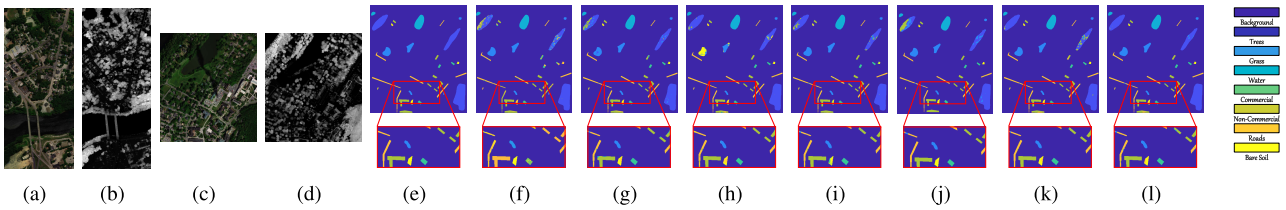



Fig. 8. Nashua–Hanover cross-domain classification maps. (a) Source HSI (Rband:54, Gband:33, Bband:15). (b) Source LiDAR. (c) Target HSI (Rband:54, Gband:33, Bband:15). (d) Target LiDAR. (e) Ground truth. (f) FrF-SSDA [25] (83.31%). (g) SCLUDA [32] (92.48%). (h) HighDAN [12] (90.65%). (i) MSDA [33] (94.39%). (j) MLUDA [31] (90.24%). (k) SASS [28] (94.28%). (l) ChirpFAN (97.16%).

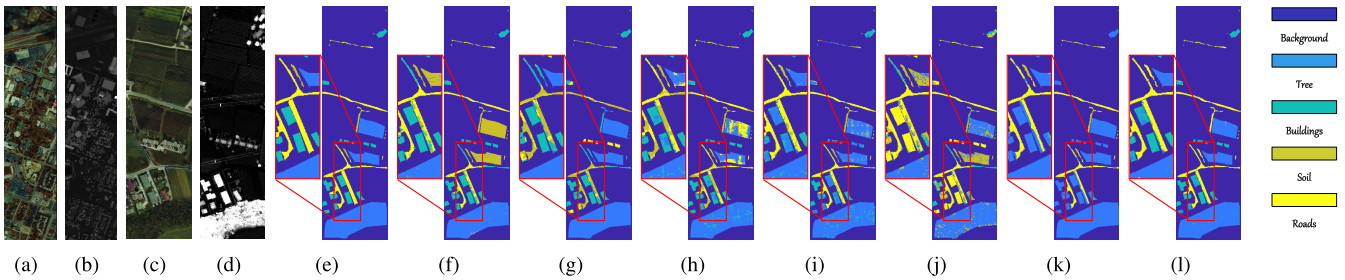



Fig. 9. Houston13–Trento cross-domain classification maps. (a) Source HSI (Rband:25, Gband:14, Bband:6). (b) Source LiDAR. (c) Target HSI (Rband:25, Gband:14, Bband:6). (d) Target LiDAR. (e) Ground truth. (f) FrF-SSDA [25] (74.43%). (g) SCLUDA [32] (82.89%). (h) HighDAN [12] (79.39%). (i) MSDA [33] (84.53%). (j) MLUDA [31] (73.69%). (k) SASS [28] (89.97%). (l) ChirpFAN (93.97%).

task. For land covers with similar spectral features, such as commercial and noncommercial areas, the proposed ChirpFAN obtains accuracies of 97.11% and 91.00%, respectively. It is a significant improvement over methods like FrF-SSDA, which score 24.61% and 71.19%. The classification of these spectrally similar classes is challenging due to disparities between the source (Houston13) and target (Houston18) domains. Even for classes that are typically difficult, ChirpFAN demonstrates robust performance. For example, grass stressed is classified with 76.94% accuracy by ChirpFAN. The improved discrimination of grass healthy and grass stressed in Table III relies on the ChirpFAN's ability to model spectral modulations differentiating these classes. The better delineation of commercial and noncommercial areas also benefits from ChirpFAN's capacity to capture spatially varying frequency patterns in both HSI and LiDAR data. The similar kinds of land covers often differ in textural complexity and structural layout, while ChirpFAN adapts to the geometric features even under domain shifts.

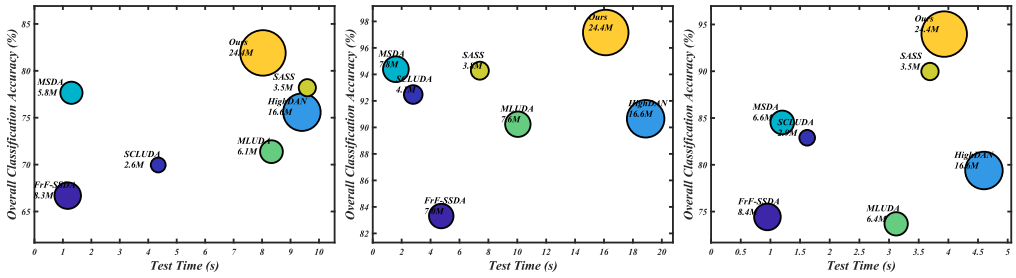
*2) Performance on the Na–Ha Scenes:* The qualitative results of competitive methods and the proposed ChirpFAN on the Na–Ha dataset are shown in Fig. 8(f)–(l), with corresponding accuracies presented in Table IV. From the classification maps, methods like FrF-SSDA with an OA of 83.31% show some misclassifications, for example, in accurately delineating bare soil from other classes like grass, leading to its 0.00% accuracy. Other methods like SCLUDA (92.48% OA) and HighDAN (90.65% OA) exhibit generally good performance but still show visual discrepancies. For instance, there is confusion between trees and commercial areas in certain parts of the HighDAN map. The proposed ChirpFAN in Fig. 8(l) shows superior performance for the various land covers, whose visual representation is cleaner with more defined boundaries for classes like roads and water. Quantitatively, as shown in Table IV, the proposed ChirpFAN achieves the highest OA of 97.16%, outperforming competitive methods on the Na–Ha cross-city classification task. Notably, for the bare soil class, ChirpFAN achieves a significantly better accuracy of 83.77% than other SOTA methods. The consistently high performance across almost all classes proves the effectiveness of the proposed ChirpFAN.

*3) Performance on the H13–T Scenes:* The qualitative results of competitive methods and the proposed ChirpFAN on the H13–T dataset are shown in Fig. 9(f)–(l), with corresponding accuracies presented in Table V. The target rural scene consists of trees, buildings, ground, and roads. From the classification maps of methods SCLUDA (82.89% OA) and HighDan (79.39% OA), there is visual confusion, such as ground appears to be misclassified as trees or buildings in HighDAN's map in Fig. 9(h). The proposed ChirpFAN in Fig. 9(l) presents a classification map with an OA of 93.97%, which shows improved differentiation between classes like buildings and ground. Quantitatively, as shown in Table V, the proposed ChirpFAN achieves the highest OA on this H13–T cross-scene classification task. For instance, it achieves 99.33% accuracy for trees and 86.50% for buildings. In this dataset, the misclassification of the ground class is obvious, for example, FrF-SSDA (1.25%) and SASS (0.00%). Due to the spatial position, similar elevation, and same composition of ground and roads in rural areas, this class is almost entirely misclassified as the classification map in Fig. 9(l) shows. Because these two classes in the rural Trento target scene share similar spectral and spatial–textural features, classification can fail even with robust domain adaptation methods when the interclass variance is low.

### F. Computational Analysis

To analyze the computational cost of the proposed ChirpFAN compared with SOTA networks, both the model sizes

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(a) (b) (c)

Fig. 10. Computation analysis of the proposed ChirpFAN on different datasets. (a) H13–H18. (b) Na–Ha. (c) H13–T.

TABLE VI
COMPUTATIONAL COST ANALYSIS ON THE H13–H18 DATASET

| Method | Params (M) | Test (s) | Training (hrs) | GPU VRAM (GB) |
|---|---|---|---|---|
| FrF-SSDA | 8.3 | 1.3 | 1.5 | 4.1 |
| SCLUDA | 2.6 | 4.4 | 2.0 | 5.3 |
| HighDAN | 16.6 | 9.6 | 3.5 | 8.2 |
| MSDA | 5.8 | 1.4 | 3.1 | 7.5 |
| MLUDA | 6.1 | 8.5 | 1.8 | 4.9 |
| SASS | 3.5 | 9.7 | 2.7 | 6.8 |
| Gabor-CNN | 19.2 | 7.6 | 4.2 | 6.8 |
| Wavelet-CNN | 18.8 | 7.9 | 4.1 | 6.5 |
| **ChirpFAN (Ours)** | **24.4** | **8.2** | **4.1** | **9.3** |

and inference time are analyzed with their performance on the three datasets used. As shown in Fig. 10, the parameter number (M) and the inference times (s) are used to evaluate the model sizes and the efficiency of methods. For all the compared methods, the same hardware and software configurations are used. As shown in Fig. 10, the model size of the proposed ChirpFAN is larger than the compared methods because of the linear matrix used in fractional frequency–phase joint extraction layers and initialization of Chirplet layers. Even though the testing time of the proposed ChirpFAN is acceptable because of the fast computations of linear transforms and stable filter design. Overall, the proposed ChirpFAN obtains the best performance with similar inference time. As shown in Table VI, while ChirpFAN is one of the largest models, its training and memory costs are comparable to other SOTA methods, and its inference time remains practical.

## VI. CONCLUSION

Aiming to overcome challenges in cross-scene classification of MSRS data, particularly domain shifts and the comprehensive extraction of features from HSIs and LiDAR data, a ChirpFAN was proposed in this article. First, a fractional feature extraction module is designed to jointly analyze spatial–elevation–frequency–phase features, providing robust representations. Second, a ChirpFA layer was integrated into a multiscale ST network (ChirpST) and then employed to extract and analyze texturally rotated and scaled patterns. Third, these components were jointly utilized within an end-to-end network with domain adaptation stages. The network not only enables robust feature extraction from MSRS data but also aligns feature distributions. Finally, experiments on three cross-scene MSRS datasets were conducted and compared with competitive methods, demonstrating their effectiveness for MSRS classification tasks. However, as computational analysis indicated a larger model size compared to some methods, future efforts will focus on exploring model compression while preserving its high accuracy. Furthermore, the model's robustness under more extreme domain shifts, where the linear chirp assumption may no longer hold (e.g., transfers across highly dissimilar sensor modalities), remains a limitation and an important direction for future investigation. Finally, the ChirpFA layer itself shows significant promise as a general-purpose replacement for standard MLP blocks in architectures designed for nonstationary signals, such as in audio, seismology, or biomedical signal processing.

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